TEACHER GUIDE DRAFT

Pearson BTEC Level 3 National Extended Diploma in Business (601/7160/1)

# Break-even

Unit 3: Personal and Business Finance
**E2 Break-even analysis**



A revision lesson. Pupils run a car dealership in firms of four and revise break-even on their own deals. Seat each firm together. Each firm answers the topic tasks together.

## The lesson

45-minute setting

| Setup | Round 1 | Round 2 | Round 3 | Final questions |
|---|---|---|---|---|

### EVERY ROUND

**Firm decisions**
3 min

→

**Deals**
4½–5 min

→

**Topic tasks**
4 min

→

**B2B deal**
2½ min
from Round 2

### LESSON SETTINGS

30 min | **45 min** | 60 min | 75 min

# What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



## CEO

Staff · Sabotage from Round 2



## Finance

Capital & Loans · Investments



## Sales & Marketing

Marketing



## Operations

Seasonal stock

# Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

## 1 Cars to break even

Round 1

They take up to 5 cars.

**Cars to break even? Round up.**

| | |
|---|---|
| Livery, one-off | **£2,400** |
| Staff training, one-off | **£2,150** |
| Customer pays per car | **£18,760** |
| You pay per car | **£16,800** |

**ANSWER**

**£3**

**(£2,400 + £2,150) ÷ £1,960 = 2.32 → 3 cars**

**Why:** Part of a car does not sell: 3 whole cars cover the costs.

## 2 Break-even limits

Round 1

You plan a fleet deal with a break-even chart.

**Which two are limits of break-even analysis?**

- **A** It leaves out the fixed costs
- **B** It cannot show the profit
- **C** It assumes one fixed price
- **D** It assumes every car is sold
- **E** It needs the firm's cash balance

**ANSWER**

**✓ It assumes one fixed price**
**✓ It assumes every car is sold**

**Why:** It assumes all output sells at one price; it does show fixed costs and profit.

## 3 Safety margin in £

Round 2

Three deals: price per car, cars sold, break-even.

**Order by margin of safety in £, lowest first?**

- **A** £15,000 a car: sell 9, break-even 6
- **B** £6,000 a car: sell 16, break-even 10
- **C** £10,000 a car: sell 12, break-even 8

**ANSWER**

1. **£6,000 a car: sell 16, break-even 10**
2. **£10,000 a car: sell 12, break-even 8**
3. **£15,000 a car: sell 9, break-even 6**

**Why:** 6 × £6,000 = £36,000, 4 × £10,000 = £40,000, 3 × £15,000 = £45,000.

# Topic tasks

## 4 Profit on the deal

Round 2

A deal with a set-up cost.

**Profit on the deal?**

| | |
|---|---|
| Livery and set-up, one-off | **£6,550** |
| Customer pays per car | **£12,475** |
| You pay per car | **£10,900** |
| They order | **9 cars** |

**ANSWER**

**£7,625**

9 × (£12,475 − £10,900) − £6,550 = £7,625

**Why:** The first cars pay the set-up; the rest is profit: £7,625.

## 5 Which price to set

Round 3

You expect 7 sales at either price. At £9,500 a car you break even at 6; at £9,000, at 8.

**Which price should you set, and why?**

**A** £9,000 a car

**B** £9,500 a car

**ANSWER**

**✓ £9,500 a car**

**Why:** At 7 sales £9,500 clears its break-even of 6; £9,000 falls short of 8.

## 6 Margin after rise

Round 3

They order 8 cars.

**Margin of safety now, in cars? Round break-even up.**

| | |
|---|---|
| Livery and set-up, one-off | **£7,900** |
| Customer pays per car | **£16,665** |
| Cost per car before the rise | **£13,150** |
| Cost rise | **10%** |

**ANSWER**

**£4**

£13,150 + 10% = £14,465; £7,900 ÷ £2,200 → 4; 8 − 4 = 4

**Why:** The cost rise ate into the margin of safety: 4 cars left.

# Final questions

Each pupil answers 6 final questions on break-even. They come from these 8 questions.

Then each pupil writes one short answer: “Analyse the risk of a low margin of safety.”

1 A component sells for £48. Variable cost is £30 a unit and fixed costs are £54,000 a month. What is monthly break-even output?

✔ **3,000 units**

2 A repair plan costs £36 a month, variable cost £12, fixed costs £72,000. Expected sales are 4,000 plans. What is the margin of safety?

✔ **1,000 plans**

3 A vehicle sells for £22,500 and its variable acquisition and preparation cost is £17,100. What is unit contribution before fixed costs?

✔ **£5,400**

4 Monthly fixed costs rise from £90,000 to £108,000 while contribution remains £300 per vehicle. By how many vehicles does break-even output increase?

✔ **60 vehicles**

5 Contribution is £2,400 a vehicle at today’s price; matching a rival cuts it to £1,500. Fixed costs are unchanged. Which effect follows?

✔ **Break-even rises: each sale covers less fixed cost**

6 A tyre service charges £120 per job, variable cost is £72 and monthly fixed costs are £38,400. What is break-even output?

✔ **800 jobs**

7 A service plan sells for £50 a month, has variable fulfilment cost of £18 and fixed monthly support cost of £96,000. What is break-even membership?

✔ **3,000 members**

8 Fixed costs are £84,000. Each vehicle sells for £16,000 and has a variable cost of £12,000. What is the break-even output?

✔ **21 vehicles**